

MODULE 1: INTRODUCTION TO OPERATIONS RESEARCH

1. Define Operations Research.

3 Marks

2024 Q1(a), 2023 Q1(a), 2025 Q2(b)

Conceptual Definition:

Operations Research (OR) is an interdisciplinary, scientific approach to executive decision-making that utilizes advanced analytical methods, mathematical modeling, statistical analysis, and algorithmic optimization to determine the best possible course of action under conditions of limited resources, technical constraints, and risk.

Standard Academic Definitions:

- **P.M. Morse & G.E. Kimball:** "Operations Research is a scientific method of providing executive departments with a quantitative basis for decisions regarding the operations under their control."
- **H.A. Taha:** "Operations Research is both an art and a science. It is an art in the sense of making intuitive judgements, and a science in the sense of establishing rigorous mathematical models."

Key Characteristics of OR:

- **System-Oriented Approach:** Evaluates how a decision impacts the entire organization rather than isolated sub-units.
- **Interdisciplinary Teamwork:** Combines expertise from mathematics, statistics, economics, business strategy, and engineering.
- **Use of Mathematical Models:** Transforms complex real-world operational problems into structured algebraic equations.
- **Optimal Decision Seeking:** Focuses on finding the single best (optimal) or near-optimal solution under specified constraints.

Real-World Corporate Example:

Amazon Global Logistics: Amazon utilizes Operations Research algorithms to evaluate real-time order delivery routes, warehouse item placement, fleet fuel consumption, and driver allocation. By modeling the fulfillment network as a mathematical optimization graph, Amazon processes millions of packages daily at minimum operational cost while meeting strict delivery time guarantees.

2. Enumerate on the Scope and Applications of Operations Research in Management.

7 Marks

2023 Q2(b), 2024 Q1(b)

Introduction & Strategic Importance:

Operations Research serves as the quantitative backbone of modern corporate decision-making. In today's hyper-competitive business environment, intuitive decision-making is insufficient. OR tools empower senior managers to allocate scarce capital, labor, raw materials, and capacity in a manner that maximizes profitability and operational resilience.

Detailed Applications Across Management Functions:

• 1. Financial Management & Capital Allocation:

- **Portfolio Management:** Using Linear Programming and quadratic models to select optimal combinations of stocks, bonds, and treasury assets that maximize expected yield while adhering to strict risk tolerance limits.
- **Capital Budgeting:** Evaluating long-term capital expansion projects and equipment replacement cycles under financial constraints.
- **Credit Risk & Cash Deployment:** Determining optimum cash balances in bank branches and ATMs to prevent cash-out scenarios while minimizing uninvested cash interest losses.

• 2. Marketing & Commercial Strategy:

- **Media Allocation Models:** Allocating fixed advertising budgets across television, digital media, print, and billboard platforms to achieve maximum audience reach and conversion.
- **Product Line Pricing & Launch Timing:** Applying Game Theory to predict competitor price reactions and establish optimal product price points during market entry.
- **Sales Territory Planning:** Grouping sales territories mathematically to minimize sales force travel distance and balance revenue potential.

• 3. Operations & Production Management:

- **Production Planning & Blending:** Determining optimal product mix targets subject to machine hours, labor shifts, and raw material availability.
- **Facility Location & Layout:** Utilizing Transportation and Facility Location models to decide where to build manufacturing plants and regional distribution hubs.
- **Inventory Control:** Balancing holding costs against setup and stock-out costs using Economic Order Quantity (EOQ) and dynamic scheduling models.

• 4. Supply Chain & Logistics:

- **Fleet Scheduling & Route Optimization:** Solving Traveling Salesman and Vehicle Routing problems to optimize multi-stop delivery routes for logistics carriers.
- **Warehouse Allocation:** Applying Assignment models to place fast-moving goods in high-accessibility warehouse zones to reduce picker travel time.

• 5. Human Resource Management:

- **Workforce Shift Scheduling:** Creating optimal shift rosters for hospital nurses, airline cabin crews, and call center agents to ensure 24/7 service coverage at minimum overtime pay.
- **Task Assignment:** Applying the Hungarian Assignment Method to match employee skills with job demands to minimize overall execution time.

Real-World Corporate Example:

Axis Bank ATM Network: Axis Bank employs OR decision models to manage cash replenishment across thousands of nationwide ATMs. The model analyzes historical withdrawal trends, local festival calendars, transit costs, and vault holding rates to determine the exact day and amount of cash to load into each machine, significantly reducing cash holding costs while preventing ATM outages.

3. Give a Brief Historical Development of Operations Research.

3 Marks

2025 Q1(a)

Historical Evolution & Timeline:

The formal origins of Operations Research date back to the late 1930s during the build-up to World War II:

- **World War II Era (1939–1945):** British military leadership convened interdisciplinary teams of mathematicians, physicists, and engineers (famously known as 'Blackett's Circus' led by Professor P.M.S. Blackett) to analyze operational problems. They mathematically optimized early warning radar systems, anti-submarine depth-charge settings, convoy formation sizes, and aerial bombing strategies.
- **Post-War Industrial Adoption (1950s):** Following WWII, scientists recognized that the mathematical models developed for military logistics could solve massive post-war industrial recovery challenges. Corporations in steel, oil refining, and transportation adopted OR to rebuild supply chains.
- **Algorithmic Breakthroughs (1947):** George B. Dantzig developed the **Simplex Algorithm** in 1947, providing a reliable method to solve Linear Programming Problems. This discovery, combined with the advent of commercial computers, launched OR into corporate management.
- **Modern Digital Era:** Today, OR is deeply integrated with computer science, Big Data, and Artificial Intelligence, powering automated real-time decision systems in global supply chains, e-commerce, and digital finance.

Historical Example:

The Royal Air Force (RAF) Radar Deployment: During the Battle of Britain, RAF scientists used OR to integrate individual radar stations into a unified defense network. This allowed the RAF to scramble fighter jets to precise interception coordinates, effectively neutralizing numerically superior enemy aircraft forces.

4. Define Operations Research and Explain the Different Phases of an Operations Research Study. 7 Marks 2025 Q2(b)

Definition:

Operations Research is a systematic, quantitative methodology that formulates real-world management problems as mathematical models to derive optimal decision policies under resource constraints.

The Six Systematic Phases of an OR Study:

1. 1. Problem Definition Phase:

The analyst collaborates with executive management to isolate the root problem. This involves establishing clear organizational objectives (e.g., maximize profit, minimize customer waiting time), identifying key decision variables, defining physical or financial constraints, and establishing boundary conditions.

2. 2. Mathematical Model Construction Phase:

The operational problem is expressed in formal mathematical notation. The model comprises:

- **Decision Variables:** Unknown quantities to be solved (e.g., x_1 , x_2).
- **Objective Function:** Algebraic equation representing profit maximization or cost minimization.
- **Constraints:** Inequality equations representing limited resources (e.g., labor hours, raw material limits).

3. 3. Model Solution Derivation Phase:

The mathematical model is solved using quantitative techniques such as the Simplex algorithm, Graphical method, Hungarian method, or computer simulations to identify optimal parameter values.

4. 4. Model Testing and Validation Phase:

Before real-world deployment, the solution is tested using historical operational data. Analysts verify whether the model accurately predicts operational performance and whether the solution remains stable under minor variations in input parameters (Sensitivity Analysis).

5. 5. Establishing Operational Controls Phase:

Control mechanisms are established to track real-world input variables. If market conditions change significantly (e.g., sudden raw material price spike), the control system alerts management to recalculate model parameters.

6. 6. Implementation Phase:

The mathematically derived solution is translated into daily operating procedures, user-friendly software interfaces, and executive guidelines for operational staff.

Real-World Corporate Example:

Commercial Airline Fleet Scheduling: An international airline follows these 6 phases to configure flight schedules and crew rosters post-pandemic. The team formulates flight schedules (Phase 1), builds mathematical programming models (Phase 2), solves them via optimization solvers (Phase 3), validates models against weather delay histories (Phase 4), sets up live tracking controls (Phase 5), and deploys the schedule to gate managers (Phase 6).

1. Give Applications of Linear Programming Problems in Management.

7 Marks

2024 Q1(b), 2023

Q2(a)

Definition & Core Concept:

Linear Programming Problem (LPP) is a mathematical optimization technique used to allocate limited resources among competing activities in an optimal manner. The word 'linear' implies that all mathematical relationships in the objective function and constraints are strictly directly proportional.

Key Managerial Applications of LPP:

- **1. Product Mix Optimization:** Manufacturing firms use LPP to determine the exact quantity of each product line to produce daily. The model maximizes net profit while ensuring consumption does not exceed available machine hours, skilled labor capacity, or raw material quotas.
- **2. Financial Investment Portfolio Selection:** Investment fund managers apply LPP to allocate capital across government securities, blue-chip equities, corporate bonds, and money market instruments. The objective maximizes annual return subject to regulatory diversification limits, liquidity requirements, and maximum risk exposure indices.
- **3. Blend Problems (Chemical & Food Processing):** Oil refineries, agricultural feed producers, and metal foundries use LPP to mix raw ingredients. For example, animal feed producers determine the cheapest combination of corn, soy, and wheat that meets strict minimum nutritional (protein, vitamin) requirements.
- **4. Transportation & Supply Chain Allocation:** Distributing finished goods from multiple factories to regional warehouses at minimum shipping cost while satisfying warehouse demands and factory capacity limits.
- **5. Advertising Media Selection:** Marketing departments allocate fixed promotional budgets across television, social media, radio, and print platforms to maximize target audience reach subject to frequency constraints.
- **6. Production Smoothing & Workforce Scheduling:** Scheduling seasonal production and temporary worker hiring across quarters to meet fluctuating market demand at minimum hiring, firing, and inventory holding costs.

Real-World Corporate Example:

Amul Dairy Processing: Amul uses Linear Programming to determine the daily allocation of collected raw milk into distinct product categories—fluid pasteurized milk, butter, cheese, milk powder, and ice cream. The LPP model maximizes total revenue while accounting for fat/snf (solid-not-fat) content, plant processing limits, shelf-life constraints, and regional consumer demand forecasts.

2. What are the Limitations in Linear Programming Problems (LPP)?

3 Marks

2024 Q2(a)

Core Limitations of LPP:

While Linear Programming is a powerful managerial tool, its practical implementation is subject to strict mathematical assumptions that may not always reflect complex business realities:

- **1. Linearity Assumption:** LPP assumes direct proportionality in costs, returns, and resource usage. In reality, business operations experience non-linear phenomena such as economies of scale, quantity bulk discounts, and diminishing marginal returns.
- **2. Certainty & Deterministic Parameters:** LPP assumes that all coefficient parameters (e.g., unit profit, resource consumption rates, raw material availability) are known with 100% certainty. Real-world business environments are subject to price volatility, demand surges, and market inflation.
- **3. Divisibility Assumption (Fractional Values):** LPP assumes decision variables can take fractional values (e.g., $x_1 = 3.75$ units). In discrete manufacturing (e.g., assembling cars, heavy machinery, or aircraft), producing fractional units is impossible, requiring complex Integer Programming adjustments.
- **4. Single Objective Focus:** Standard LPP optimizes a single objective function (either maximize profit or minimize cost). Modern organizations frequently pursue multi-objective targets, such as simultaneously maximizing market share, maintaining employment stability, and reducing carbon emissions.
- **5. Static Model Nature:** LPP evaluates operational choices at a single point in time, whereas corporate strategic decisions are dynamic and evolve across multiple consecutive financial quarters.

Practical Business Example:

Automobile Assembly Line: Standard LPP might calculate an optimal production target of 14.6 luxury trucks per day. Because a factory cannot ship 0.6 of a truck, rounding off decision variables can violate resource constraints or yield sub-optimal revenue, illustrating the divisibility limitation.

3. What are the Limitations of the Graphical Approach to Solve LPP?

3 Marks

2024 Q5(a)

Key Limitations of the Graphical Method:

The Graphical Method is a visual technique used to solve Linear Programming problems by plotting constraint lines on a Cartesian coordinate plane and identifying the feasible region corner points. Its application is severely restricted by the following factors:

- **1. Strict Two-Variable Limit:** Graphical analysis is practically limited to problems containing only **two decision variables** (x_1, x_2). While three variables can theoretically be modeled in 3D space, drawing and evaluating 3D planes manually is highly complex. For real-world business problems containing dozens or hundreds of decision variables, the graphical method is completely unusable (requiring the Simplex algorithm).
- **2. Drafting & Coordinate Inaccuracy:** Graphical solutions depend on accurate manual plotting of linear equations, axis scales, and slope angles. Minor drafting errors can lead to incorrectly plotted corner points, resulting in wrong optimal values.
- **3. Visually Overburdened Feasible Regions:** When an LPP contains numerous inequality constraints (e.g., 10 or 15 constraint lines), the graph becomes visually cluttered. Identifying the precise bounded feasible polygon and its extreme corner points becomes extremely difficult.
- **4. Inefficiency in Sensitivity Analysis:** Evaluating how the optimal solution changes when resource limits or profit coefficients fluctuate (sensitivity analysis) requires redrawing the entire graph repeatedly, making it inefficient compared to matrix-based computer solvers.

Practical Business Example:

Retail Store Inventory Optimization: A retail store manager deciding stock levels for 50 distinct consumer product items cannot utilize the graphical method due to the 50-variable dimension. They must rely on computational Simplex algorithms instead.

1. Write Down the Different Steps in the Decision-Making Process.

7 Marks

2024 Q2(b), 2023 Q6(b)

Introduction to Decision Theory:

Decision Theory provides a logical, quantitative framework for making optimal managerial choices when facing alternative courses of action, uncertain future environments, and varying financial payoffs.

Systematic Steps in the Decision-Making Process:

1. Step 1: Problem Identification & Objective Setting:

Clearly state the business decision required. Define the organizational goal, such as maximizing net financial profit, minimizing capital risk, or maximizing market share.

2. Step 2: Identification of Action Alternatives (Strategies):

List all available decisions under executive control. These controlled options are called **Strategies or Alternatives** ($A_1, A_2, A_3, \dots, A_m$).

3. Step 3: Identification of States of Nature (Future Events):

Identify all possible future environmental conditions or market states that can occur. These external events are outside executive control and are termed **States of Nature** ($S_1, S_2, S_3, \dots, S_n$). Examples include economic boom, recession, or stable market conditions.

4. Step 4: Construction of the Payoff Matrix:

Construct a grid table detailing the conditional financial outcome (profit or cost) resulting from each combination of action alternative (A_i) and state of nature (S_j). The entry P_{ij} represents the payoff for adopting Strategy A_i when State of Nature S_j occurs.

5. Step 5: Assessment of Probabilities (If Applicable):

Under decision-making under risk, assign objective or subjective probabilities ($P(S_j)$) to each state of nature, ensuring $\sum P(S_j) = 1.0$.

6. Step 6: Selection and Application of Decision Criterion:

Apply the appropriate decision model based on the information environment:

- *Under Uncertainty*: Maximax, Maximin, Minimax Regret, or Laplace criteria.
- *Under Risk*: Expected Monetary Value (EMV) or Expected Opportunity Loss (EOL).

7. Step 7: Selection and Implementation of Optimal Course of Action:

Select the strategy yielding the highest optimal expected reward and translate it into executive operating directives.

Real-World Corporate Example:

Real Estate Development Firm: A property developer decides whether to build a large shopping mall (A_1), a residential complex (A_2), or hold cash (A_3). They identify potential market states: high economic growth (S_1), moderate growth (S_2), or recession (S_3). They build a payoff matrix mapping projected net cash flows, calculate expected outcomes, apply the Maximin criterion to mitigate recession risks, and select the residential complex project.

2. State the Basic Assumptions Underlying Job Sequencing Problems.

3 Marks

2024 Q7(a), 2025

Q2(a)

Core Assumptions in Sequencing Models:

Job Sequencing involves determining the optimal order in which n jobs should be processed across m machines to minimize total elapsed processing time (T) and machine idle times. Standard sequencing algorithms (such as Johnson's Rule) rely on the following explicit operational assumptions:

- **1. Constant & Known Processing Times:** Processing time for each job on every machine is fully known, fixed, and independent of the order in which jobs are executed.
- **2. Strict & Pre-determined Machine Routing Order:** Each job follows a specified, invariant sequence through the processing units (e.g., Job must go to Machine A first, then Machine B , then Machine C).
- **3. Single-Task Machine Availability:** Each machine can process only one job at a time. Simultaneous parallel job processing on a single machine is not permitted.
- **4. Non-Preemption Operational Policy:** Once a job starts processing on a machine, it must complete its full operation duration without interruption. Pausing a job to process another is not allowed.
- **5. Negligible Setup & Transport Times:** Machine setup times, tooling changeover delays, and material transport times between machines are negligible or included in the estimated job processing times.
- **6. Static Job Pool:** All n jobs are available for processing at the beginning of the scheduling time horizon ($t = 0$). No new jobs arrive mid-process.

Practical Industrial Example:

Book Publishing House: Printing 5 distinct book orders ($n = 5$) through 3 sequential operations ($m = 3$: Printing $\rightarrow$ Binding $\rightarrow$ Laminating). The scheduler assumes paper rolls are pre-loaded, machines process one book batch at a time without breaking down, and transportation between machines takes negligible time.

3. Enumerate on the Different Criteria of Decision Making Under Uncertainty. 7 Marks

2023 Q5(b)

Introduction to Decision Making Under Uncertainty:

Decision making under uncertainty occurs when a manager faces multiple states of nature but lacks historical data or objective evidence to assign probabilities to their occurrence. In such situations, the decision selection depends on the executive's attitude toward risk (optimistic, pessimistic, or risk-averse).

The Five Classical Criteria Under Uncertainty:

- **1. Maximax Criterion (Criterion of Optimism):**

The decision-maker is purely optimistic. They locate the maximum possible payoff for each alternative and select the strategy that offers the ultimate maximum outcome ("best of the best").

Formula: $\text{Select } \max_i [\max_j P_{ij}]$

- **2. Maximin Criterion (Criterion of Pessimism / Wald's Criterion):**

The decision-maker is completely conservative and risk-averse. They assume nature will present the worst-case scenario. They identify the minimum payoff for each strategy and choose the alternative that maximizes this minimum return ("best of the worst").

Formula: $\text{Select } \max_i [\min_j P_{ij}]$

- **3. Minimax Regret Criterion (Savage's Opportunity Loss Criterion):**

Focuses on minimizing maximum potential decision regret. A Regret (Opportunity Loss) Matrix is created by subtracting each payoff from the maximum payoff under that state of nature ($R_{ij} = \max_k P_{kj} - P_{ij}$). The manager finds the maximum regret for each strategy and chooses the one with the smallest maximum regret.

Formula: $\text{Select } \min_i [\max_j R_{ij}]$

- **4. Laplace Criterion (Criterion of Equal Likelihood / Equal Probability):**

Assumes that because state probabilities are completely unknown, every state of nature has an equal probability of occurrence ($P(S_j) = 1/n$). The manager calculates the simple average payoff for each alternative and chooses the strategy with the highest average return.

- **5. Hurwicz Alpha Criterion (Criterion of Realism / Compromise):**

Strikes a balance between extreme optimism and pessimism using a Coefficient of Optimism (α), where $0 \leq \alpha \leq 1$. The weighted payoff is calculated as:

$\text{Payoff} = \alpha \cdot (\text{Max Payoff}) + (1 - \alpha) \cdot (\text{Min Payoff})$

The decision-maker selects the alternative with the highest calculated weighted payoff.

Real-World Corporate Example:

New Tech Product Commercialization: A consumer electronics firm launching an innovative smartwatch in an untested market evaluates 3 plant capacity options (Small, Medium, Large). Lacking market probability data, a risk-averse CFO applies the Maximin criterion to select the Small Plant capacity, guaranteeing minimal financial losses if consumer demand fails to materialize.

4. Why is Job Sequencing Important? / Define the Term Sequencing and State Its Importance.

3 Marks

2023 Q6(a), 2025 Q6(a)

Definition & Core Concept:

Job Sequencing is a systematic operational scheduling technique used to determine the precise order in which a given set of manufacturing or service jobs should be processed across shared production machines or workstations.

Managerial Importance of Job Sequencing:

- **1. Minimization of Total Elapsed Time (\$T\$):** Proper sequencing compresses the total duration required to complete an entire batch of jobs from start to finish, accelerating order turnaround times.
- **2. Reduction of Machine Idle Times:** Prevents bottleneck workstations from standing idle while waiting for preceding processing steps, maximizing equipment utilization.
- **3. Minimization of Work-in-Progress (WIP) Inventory:** Systematic job flow prevents semi-finished inventory from piling up between machine stations, reducing floor space congestion and working capital holding costs.
- **4. Meeting Customer Delivery Deadlines:** Minimizes tardiness and delivery delays, improving customer satisfaction and avoiding contractual penalty fees.
- **5. Optimization of Resource Efficiency:** Balances workloads across machinery, improving operational throughput without purchasing additional equipment.

Practical Industrial Example:

Automobile Repair Workshop: A garage servicing 10 vehicles across washing, mechanical repair, and paint booth bays uses sequencing algorithms to order vehicles, minimizing total customer wait times and keeping service bays operating at peak capacity.

5. Write a Short Note on Decision Theory.

3 Marks

2025 Q5(a)

Overview & Purpose of Decision Theory:

Decision Theory is a body of quantitative and analytical concepts designed to assist business executives in selecting optimal courses of action when facing complex, uncertain operational environments.

Core Decision Framework Environments:

- **1. Decision Making Under Certainty:** The decision-maker has complete, accurate knowledge of future events. Every action leads to a single, guaranteed outcome (e.g., investing in fixed government bonds).
- **2. Decision Making Under Risk:** Future states of nature are uncertain, but historical data or market research allows managers to assign objective or subjective probabilities to outcomes (analyzed via Expected Monetary Value - EMV).
- **3. Decision Making Under Uncertainty:** Future outcomes are completely unknown, and reliable probabilities cannot be established. Choices are evaluated using criteria such as Maximax, Maximin, or Laplace.
- **4. Decision Making Under Conflict (Game Theory):** The outcome depends on decisions made by rational, active competitors pursuing opposing objectives.

Practical Executive Example:

Pharmaceutical R&D Investment: A pharma executive uses decision trees (a core Decision Theory tool) to evaluate whether to invest \$50 million in Phase III clinical trials for a new drug, weighing clinical success probabilities against market returns.

1. What is an Unbalanced Case in a Transportation Model? / What is Meant by an Unbalanced Transportation Problem? 3 Marks 2024 Q3(a), 2025 Q3(a)

Conceptual Definition:

An **Unbalanced Transportation Problem** is a logistics optimization problem in which the Total Available Supply at all source origins is not equal to the Total Required Demand across all destination warehouses.

Mathematical Condition:

$\sum_{i=1}^m a_i \neq \sum_{j=1}^n b_j$ (where a_i = supply at origin i , and b_j = demand at destination j)

Method of Resolution:

Standard transportation algorithms (such as NWCM, VAM, and MODI) require a balanced system ($\sum \text{Total Supply} = \sum \text{Total Demand}$). To resolve an unbalanced problem, a fictitious source or destination is introduced:

- **Case 1: Total Supply > Total Demand ($\sum a_i > \sum b_j$):**
Introduce a **Dummy Destination Column** with a demand equal to the excess supply ($\sum a_i - \sum b_j$). Set all transportation unit cost values in this dummy column to zero ($c_{i, \text{dummy}} = 0$).
- **Case 2: Total Demand > Total Supply ($\sum b_j > \sum a_i$):**
Introduce a **Dummy Origin Row** with a supply equal to the deficit ($\sum b_j - \sum a_i$). Set all transportation unit cost values in this dummy row to zero ($c_{\text{dummy}, j} = 0$).

Practical Logistics Example:

Cement Manufacturing & Regional Depots: A cement manufacturer has 3 plants producing 500 tons total, while regional depots request 650 tons. The analyst adds a Dummy Plant Row supplying 150 tons at zero shipping cost, balancing the model before applying Vogel's Approximation Method.

2. What is Degeneracy in Transportation Problems? / Write a Note on Degeneracy of Transportation Problem. 3 Marks 2023 Q7(a), 2025 Q4(a)

Conceptual Definition:

Degeneracy in a Transportation Problem occurs when the total number of occupied (allocated) basic cells is strictly less than $m + n - 1$, where m is the number of origins (rows) and n is the number of destinations (columns).

Operational Impact:

When a transportation solution is degenerate, evaluating shadow prices (u_i and v_j values) and opportunity costs ($\Delta_{ij} = c_{ij} - (u_i + v_j)$) during optimality testing via the MODI (Modified Distribution) Method becomes impossible because there are insufficient allocations to establish a closed-loop network.

Types of Degeneracy & Resolution Methods:

- **1. Degeneracy at Initial Stage:** Occurs during initial basic feasible solution generation when a single allocation simultaneously satisfies both row supply and column demand.

Resolution: Place an infinitely small positive quantity, denoted as **Epsilon (ϵ)**, into an unallocated cell that has a low unit shipping cost and does not form a closed loop with existing basic allocations.

- **2. Degeneracy at Intermediate Optimality Stage:** Occurs during MODI iterations when shifting allocations along a closed loop causes two or more occupied cells to hit zero simultaneously.

Resolution: Retain one of the newly vacated zero-allocation cells as a basic cell assigned a quantity of ϵ .

Practical Operational Example:

FMCG Shipping Grid: In a 3-origin, 4-destination transportation matrix ($3 + 4 - 1 = 6$ required allocations), an initial VAM solution yields only 5 allocated cells. The analyst places ϵ in an empty cell with a low shipping cost to restore 6 basic cells, enabling MODI optimality testing.

3. Explain the Unbalanced Case and Degeneracy in Transportation Problems in Detail.

7 Marks

2023 Q7(a), 2024 Q3(a), 2025 Q3(a)/Q4(a)

Introduction:

Transportation models optimize the physical movement of commodities from supply origins to demand destinations. In real-world distribution networks, operational complexities such as supply-demand imbalances and degenerate mathematical loops frequently arise.

Section 1: The Unbalanced Transportation Problem:

Standard transportation algorithms require mathematical balance ($\sum a_i = \sum b_j$). When supply and demand are unequal, the model is adjusted using dummy variables:

- **Supply Surplus ($\sum \text{Supply} > \sum \text{Demand}$):**

A dummy column representing an imaginary warehouse is added to absorb surplus output. The unit shipping cost to this dummy destination is set to \$0\$, representing inventory retained at the factory warehouse.

- **Demand Deficit ($\sum \text{Demand} > \sum \text{Supply}$):**

A dummy row representing an imaginary plant is added to fulfill the demand shortfall. The unit shipping cost from this dummy origin is set to \$0\$, representing unfulfilled customer orders.

Section 2: Degeneracy in Transportation Problems:

A basic feasible solution to an $m \times n$ transportation problem must satisfy two mathematical conditions:

1. Number of basic allocations must equal $m + n - 1$.
2. The basic allocated cells must occupy **independent positions** (contain no closed loops).

If allocations fall below $m + n - 1$, the problem is degenerate. Degeneracy is resolved by allocating an arbitrarily small value ϵ (epsilon) to an unallocated cell, where $\epsilon \rightarrow 0$. ϵ follows these arithmetic rules:

- $x + \epsilon = x$
- $x - \epsilon = x$
- $\epsilon > 0$

Real-World Supply Chain Example:

Petroleum Refinery Logistics: Bharat Petroleum routes fuel from 3 refineries to 5 distribution hubs. If total refinery production is 1,200 kiloliters while hub demand is 1,500 kiloliters, BP adds a Dummy Refinery Row providing 300 kiloliters at zero transit cost. If the resulting initial VAM solution has fewer than $3 + 5 - 1 = 7$ allocations, BP places ϵ in a low-cost unallocated route to perform MODI optimization.

1. What is a Saddle Point in Game Theory?

3 Marks

2024 Q4(a)

Conceptual Definition:

A **Saddle Point** in a zero-sum game is a position (cell) in the payoff matrix where the maximum of the row minimums (Maximin) equals the minimum of the column maximums (Minimax).

Mathematical Condition:

$$\max\{\text{Maximin}\} = \min\{\text{Minimax}\} = v \quad (\text{Value of the Game})$$

Key Properties of a Saddle Point:

- **Equilibrium Strategy Point:** Represents a stable outcome where neither player has an incentive to unilaterally alter their strategy.
- **Pure Strategy Indicator:** The presence of a saddle point indicates that the game can be solved using **Pure Strategies**, meaning both players will select a specific, fixed action 100% of the time.
- **Saddle Point Location Method:**
 1. Find the minimum value in each row and select the maximum of these minimums (**Maximin**).
 2. Find the maximum value in each column and select the minimum of these maximums (**Minimax**).
 3. If $\max\{\text{Maximin}\} = \min\{\text{Minimax}\}$, that cell is the Saddle Point, and its payoff value is the Value of the Game (v).

Practical Market Example:

Duopoly Pricing Competition: Two competing telecom companies (A and B) evaluate price discounts. If Firm A's Maximin strategy return equals Firm B's Minimax loss at a payoff of \$0, a Saddle Point exists, and both firms maintain standard pricing without entering price wars.

2. What is an Assignment Problem?

3 Marks

2023 Q3(a)

Conceptual Definition:

An **Assignment Problem** is a specialized sub-class of Linear Programming and Transportation problems in which the objective is to assign a set of n tasks (jobs, projects, or routes) to n resources (workers, machines, or vehicles) on a strict **one-to-one basis** at minimum total cost or minimum execution time.

Key Structural Characteristics:

- **Square Cost Matrix ($n \times n$):** The number of tasks must equal the number of available resources.
- **One-to-One Allocation Restriction:** Exactly one resource is assigned to exactly one task (e.g., worker i gets job j).
- **Binary Decision Variable:** The decision variable x_{ij} takes on binary values:
 $x_{ij} = 1$ (if resource i is assigned to task j), $x_{ij} = 0$ (otherwise)
- **Solution Method:** Solved efficiently using the **Hungarian Method** (Kuhn-Munkres algorithm) rather than standard transportation methods, avoiding degenerate solutions.

Practical Corporate Example:

IT Consulting Firm: Assigning 4 software consultants to 4 client software projects. Because consultants have varying skill sets and billing rates, management applies the Hungarian Assignment Method to minimize total project delivery cost.

3. Explain the Max. Min and Min. Max Principles with an Example for Each. 7 Marks 2025 Q6(b)

Introduction to Game Principles:

In two-person zero-sum games, two rational competing players adopt opposing strategic goals. The Maximin and Minimax principles establish optimal strategies when competitors aim to protect their interests against worst-case scenarios.

1. The Maximin Principle (Used by Maximizing Player A):

Player \$A\$ wishes to maximize their financial payoff. Recognizing that competing Player \$B\$ will attempt to minimize Player \$A\$'s gain, Player \$A\$ acts conservatively:

- Player \$A\$ identifies the **minimum possible payoff** for each of their available rows.
- Player \$A\$ then selects the strategy that yields the **maximum among these minimum outcomes**.
- *Objective:* Guarantees that Player \$A\$'s gain will not fall below this calculated value, regardless of Player \$B\$'s actions.

2. The Minimax Principle (Used by Minimizing Player B):

Player \$B\$ incurs costs equal to Player \$A\$'s gains. Player \$B\$ aims to minimize their overall payout:

- Player \$B\$ identifies the **maximum potential payout** in each of their available columns.
- Player \$B\$ then selects the strategy that yields the **minimum among these maximum payouts**.
- *Objective:* Ensures Player \$B\$'s financial payout will not exceed this calculated limit, regardless of Player \$A\$'s actions.

Illustrative Matrix Example:

Consider the following \$2 \times 2\$ Payoff Matrix for Player \$A\$ (Row Player) and Player \$B\$ (Column Player):

Player A \ Player B Strategy	Strategy B ₁	Strategy B ₂	Row Minimum
Strategy A ₁	6	2	2
Strategy A ₂	8	4	4 ← ext{Maximin}
Column Maximum	8	4 ← ext{Minimax}	Value = 4

Interpretation:

- **Player A's Maximin Strategy:** $\max(2, 4) = 4$ (Selects Strategy \$A_2\$). Player \$A\$ guarantees a profit of at least 4.
- **Player B's Minimax Strategy:** $\min(8, 4) = 4$ (Selects Strategy \$B_2\$). Player \$B\$ limits their payout to at most 4.
- Because $\text{ext{Maximin}} = \text{ext{Minimax}} = 4$, a **Saddle Point** exists at position \$(A_2, B_2)\$, and the Value of the Game is \$4\$.

Real-World Corporate Example:
Commercial Airline Pricing War: IndiGo (\$A\$) and Air India (\$B\$) select ticket pricing structures during peak holiday seasons. IndiGo uses Maximin to protect base revenue, while Air India uses Minimax to limit market share losses to IndiGo.

4. Define Key Terms in Game Theory: (i) Payoff Matrix, (ii) Pure Strategy, (iii) Mixed Strategy, (iv) Value of Game, (v) Fair Game. 10 Marks 2025 Q5(c)

Comprehensive Definitions & Conceptual Breakdown:

1. (i) Payoff Matrix:

A **Payoff Matrix** is a structured table that summarizes the conditional gains or losses resulting from combinations of strategy choices between competing players in a game. For a two-person game, Row Player A 's strategies are plotted along rows, while Column Player B 's strategies are plotted along columns. Positive cell entries represent gains to Row Player A (and losses to Column Player B), while negative entries represent losses to Row Player A .

2. (ii) Pure Strategy:

A **Pure Strategy** is a deterministic decision policy in which a player chooses the exact same action with 100% probability every time the game is played ($p_i = 1.0$). A pure strategy exists when the payoff matrix contains a stable Saddle Point ($\text{ext}\{\text{Maximin}\} = \text{ext}\{\text{Minimax}\}$). Players do not need to alter their strategies or keep their opponent guessing.

3. (iii) Mixed Strategy:

A **Mixed Strategy** is a stochastic decision policy adopted when no single Saddle Point exists in the payoff matrix. Rather than choosing one fixed action, a player selects from available strategies according to calculated probabilities ($0 < p_i < 1$, where $\sum p_i = 1.0$). This keeps the competitor guessing and maximizes expected return over repeated plays.

4. (iv) Value of the Game (v):

The **Value of the Game** (v) represents the expected net financial outcome or average payout per play when both competing players adopt their optimal strategies (pure or mixed). If Player A earns v , Player B loses v in a zero-sum game.

5. (v) Fair Game:

A game is defined as a **Fair Game** if the calculated Value of the Game is exactly equal to zero ($v = 0$). In a fair game, neither competing player enjoys a built-in strategic advantage; over repeated plays, expected net gains and losses balance out to zero.

Real-World Corporate Example:

Fast-Food Marketing War: McDonald's (A) and Burger King (B) allocate promotional ad budgets across TV, digital, and print channels. The Payoff Matrix maps market share shifts. Lacking a Saddle Point, both brands adopt a **Mixed Strategy** (e.g., McDonald's splits ad budget 60% TV, 40% Digital) to keep the competitor from counter-programming effectively.

MODULE 6: PROJECT MANAGEMENT (PERT & CPM NETWORKS)

1. Distinguish Between PERT and CPM Methods.

7 Marks

Repeated in ALL 3 Years (2024 Q7(b), 2023 Q4(b), 2025 Q7(b))

Introduction to Project Management Networks:

PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method) are the two foundational network analysis techniques used to plan, schedule, monitor, and control complex projects.

Detailed Comparative Analysis Table:

Comparison Feature	PERT (Program Evaluation & Review Technique)	CPM (Critical Path Method)
1. Core Nature & Approach	Probabilistic model. Designed for projects with uncertain, unpredictable activity durations.	Deterministic model. Designed for projects with known, predictable activity durations.
2. Fundamental Focus	Event-oriented technique. Emphasizes achieving key project milestones and event completion.	Activity-oriented technique. Emphasizes task execution and resource allocation across activities.
3. Time Estimates Used	Utilizes Three Time Estimates for each activity: Optimistic (t_o), Most Likely (t_m), and Pessimistic (t_p).	Utilizes a Single Time Estimate based on historical operational data.
4. Industry Application	Used for non-repetitive, high-uncertainty projects such as R&D, space exploration, and new software development.	Used for repetitive, standardized projects such as civil construction, plant maintenance, and shipbuilding.
5. Cost & Crashing Focus	Primarily focused on timeline probabilities; does not directly incorporate activity crashing cost trade-offs.	Explicitly incorporates Time-Cost Trade-offs (Crashing) to reduce project completion time at minimum crash cost.
6. Mathematical Model	Uses Beta probability distribution to calculate Expected Time ($t_e = \frac{t_o + 4t_m + t_p}{6}$) and Variance ($\sigma^2 = (\frac{t_p - t_o}{6})^2$).	Uses fixed scalar activity durations to calculate Critical Path duration and Floats directly.

Real-World Comparative Example:

- PERT Application: ISRO Chandrayaan Mission:** Developing a space lander involves unmapped technologies and uncertain testing times, requiring PERT's 3-time estimate probability framework.
- CPM Application: L&T Metro Rail Construction:** Building a standardized metro line uses CPM because concrete curing and track laying times are well established from past projects.

2. What are Looping and Dangling Errors in Networking?

3 Marks

2024 Q6(a)

Conceptual Overview:

When constructing a project network diagram (AOA - Activity-on-Arrow), strict logical rules must be followed. Looping and Dangling are logical errors that invalidate the network structure.

1. Looping (Cycling) Error:

- **Definition:** A logical error where a series of activities forms an endless closed circular path (e.g., Activity A → B → C → A).
- **Impact:** Creates an infinite loop, making it mathematically impossible to perform forward and backward pass calculations or determine the project completion date.
- **Correction:** Re-examine activity precedence relationships to eliminate circular dependencies.

2. Dangling Error:

- **Definition:** A logical error where an activity path disconnects from the main network flow, leaving an intermediate event node loose without any succeeding arrow connecting it to the final project completion node.
- **Impact:** Leaves project activities hanging uncompleted, leading to incorrect critical path calculations.
- **Correction:** Every intermediate event node (except the final end node) must connect to a successor activity that leads to the final project completion node.

Practical Network Example:

In a house construction project network, if the 'Roofing Work' node does not connect to 'Interior Painting' or 'Final Handover', it creates a **Dangling Error**. If 'Roofing' points back to 'Foundation Work', it creates a **Looping Error**.

3. What Do You Understand by "Work-Breakdown Structure" (WBS)?

3 Marks

2023 Q4(a)

Conceptual Definition:

A **Work Breakdown Structure (WBS)** is a deliverable-oriented, hierarchical decomposition of a complete project into smaller, more manageable sub-projects, work packages, and individual activities.

Key Structural Levels of WBS:

- **Level 1: Total Project Scope** (e.g., Constructing an Airport Terminal).
- **Level 2: Major Sub-Projects / Deliverables** (e.g., Runway, Terminal Building, Baggage System).
- **Level 3: Work Packages** (e.g., Terminal Foundation, Electrical Wiring, HVAC Installation).
- **Level 4: Operational Activities** (e.g., Pouring concrete slab, pulling copper wiring).

Managerial Benefits of WBS:

- **1. Clear Scope Definition:** Prevents scope creep by clearly defining all project tasks.
- **2. Precise Cost & Resource Estimation:** Allows project managers to estimate budgets and assign responsibility at the individual work package level.
- **3. Foundation for Network Diagrams:** Serves as the direct input for identifying activities and drawing PERT/CPM network diagrams.

Practical Corporate Example:

ERP Software Deployment: Infosys uses WBS to decompose a massive ERP implementation into 5 sub-project phases: System Requirements → Database Architecture → Custom Coding → User Acceptance Testing → Go-Live Deployment.

4. What is the "Project Control Phase" / Describe the Phases of Project Management.

3 Marks

2023 Q5(a), 2025 Q7(a)

The Three Core Phases of Project Management:

1. **Project Planning Phase:** Defining project objectives, scope, work breakdown structures (WBS), resource availability, and activity relationships before work begins.
2. **Project Scheduling Phase:** Constructing PERT/CPM network diagrams, calculating activity start/finish times (\$ES, EF, LS, LF\$), identifying the Critical Path, and establishing deadlines.
3. **Project Control Phase:** The active operational phase where project execution is monitored against the scheduled baseline.

Key Elements of Project Control:

- **Performance Tracking:** Comparing actual task progress against scheduled timelines and cost budgets.
- **Variance Analysis:** Identifying delays on critical path activities (\$TF = 0\$).
- **Corrective Action (Resource Reallocation):** Shifting slack resources from non-critical activities (high float) to delayed critical tasks, or crashing activities to bring the project back on schedule.

Practical Execution Example:

Highway Construction Project: During the Project Control Phase, a site engineer notices that bridge pillar construction is 5 days behind schedule. Because this pillar is a critical activity, the engineer shifts cranes from a non-critical access road activity (high float) to accelerate pillar curing, keeping the project on schedule.